

RAG Application Guide

Document ingestion

A RAG application can ingest PDF, HTML, and Markdown sources. During ingestion, the application should preserve useful metadata such as source filename, file type, page number when available, and chunk identifier.

Chunk size and overlap

Chunk size and overlap should be configurable. Chunking should balance context preservation with focused retrieval. Overlap helps preserve information that crosses a chunk boundary.

Embedding model

Every chunk is converted into an embedding vector. For reproducibility, record the embedding model and the dimensionality of the produced vectors.

Vector store

A vector store keeps embeddings and metadata and supports nearest-neighbor retrieval. A local store can reduce infrastructure cost for a small or lightly queried application.

Grounded generation

The generator should answer from retrieved context and cite the chunks used. When the context is insufficient, it should explicitly decline to answer rather than hallucinate.

Operational measurements

A RAG API should record retrieval latency, total latency, retrieved chunk count, and token usage when the generator exposes token counts.